

5. **Image 5** shows surgery on a patient with heart disease caused by a faulty gene.

Image 5



(a) After surgery, it is important that wounds are sealed.
Explain why wounds must be sealed. [2]

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(b) Patients are given antibiotics after surgery.
State the name of **one** antibiotic **and** the reason why antibiotics are given to patients. [2]

Antibiotic

Reason

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(c) The surgeons are using stem cells taken from the patient to replace damaged heart muscle.
Suggest **one** benefit of using stem cells from the patient rather than from a donor. [1]

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(d) Scientific research has produced a map of all the genes present in humans.
(i) Complete the sentence below by underlining the correct term. [1]

The complete map of human genes is called the human:

genome gene pool genotype



Examiner
only

(ii) Suggest **two** ways by which scientists make their work known to other scientists around the world. [2]

I.

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II.

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